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Engineering Applications of Artificial Intelligence **October 22, 2026**

Dear Editors,

We are pleased to submit our manuscript entitled **"A Shapley-value framework for quantifying module contributions in context-augmented decision support: application to parcel-level land management"** for consideration for publication in *Engineering Applications of Artificial Intelligence*.

Following the journal's submission conditions, we state the two contributions explicitly. **The contribution in artificial intelligence** is a framework that models a modular, context-augmented architecture as a cooperative game and applies Shapley values to attribute response quality to individual data modules, combining a full factorial ablation over module coalitions, a hybrid human-in-the-loop evaluation panel, and non-parametric bootstrap confidence intervals. **The application in engineering** is a geospatial decision support system for parcel-level land management, which fuses cadastral records, satellite multispectral indices, aerial orthophotos and retrieval-augmented document search to answer natural-language queries about agricultural plots.

The work addresses a gap that affects any modular system built on augmented context: although such architectures are increasingly deployed as decision support tools, there is no established method for determining how much each information source contributes to the quality of the output. Our evaluation panel combines two human evaluators and two distinct large language model judges, scoring 143 queries over 15 parcels across five metrics. Across that dataset, the orthophoto module emerges as the dominant contributor, adding 0.9 out of 5 points (18%) to the mean score, while the external context modules together raise response utility by 32%. The retrieval module exhibits high query-dependent volatility and frequently negative contributions, indicating that retrieval should be invoked selectively rather than by default. Complementary analyses of context length and inter-judge agreement show that the system is resilient to information overload and quantify where human and machine judges diverge.

In line with the journal's scope, the results are validated on a publicly released dataset. The complete evaluation corpus, containing the queries, the parcel orthophotos, the retrieved multi-domain context and the generated responses, is deposited in a public repository under the Apache 2.0 license, so that the reported evaluation can be independently reproduced and the attribution framework reused on other modular systems. An anonymous reviewer link to the deposit is given in the Data availability statement of the manuscript; the permanent identifier, withheld from the anonymized version because it discloses author identity, has been supplied to the editorial office.

This manuscript is a substantially extended version of our conference paper presented at the *2025 4th International Conference on Geographic Information and Remote Sensing Technology (GIRST)* (DOI: 10.1109/GIRST67753.2025.11382141), which proposed the multimodal system evaluated here. The journal version shifts the focus from the technical implementation of that system to a mathematically grounded evaluation of its outputs, and contributes both the attribution methodology and the dataset built for the study. We note that the two papers share no text beyond bibliographic references and the authorship statement.

In accordance with the journal's double anonymized review policy, the manuscript has been anonymized and the title page, carrying author details, acknowledgements and the CRediT statement, is submitted as a separate file.

We confirm that this manuscript is original, has not been published elsewhere, and is not currently under consideration by any other journal. All authors have approved the manuscript and agree with its submission. The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

We thank you for considering our work and look forward to your editorial decision.

Sincerely,
Prof. J. Molleda, on behalf of all authors.